Physical Optics (polarization)

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Light is the interaction of electric and magnetic fields travelling through space. The electric and magnetic vibrations of a light wave occur perpendicularly to each other.

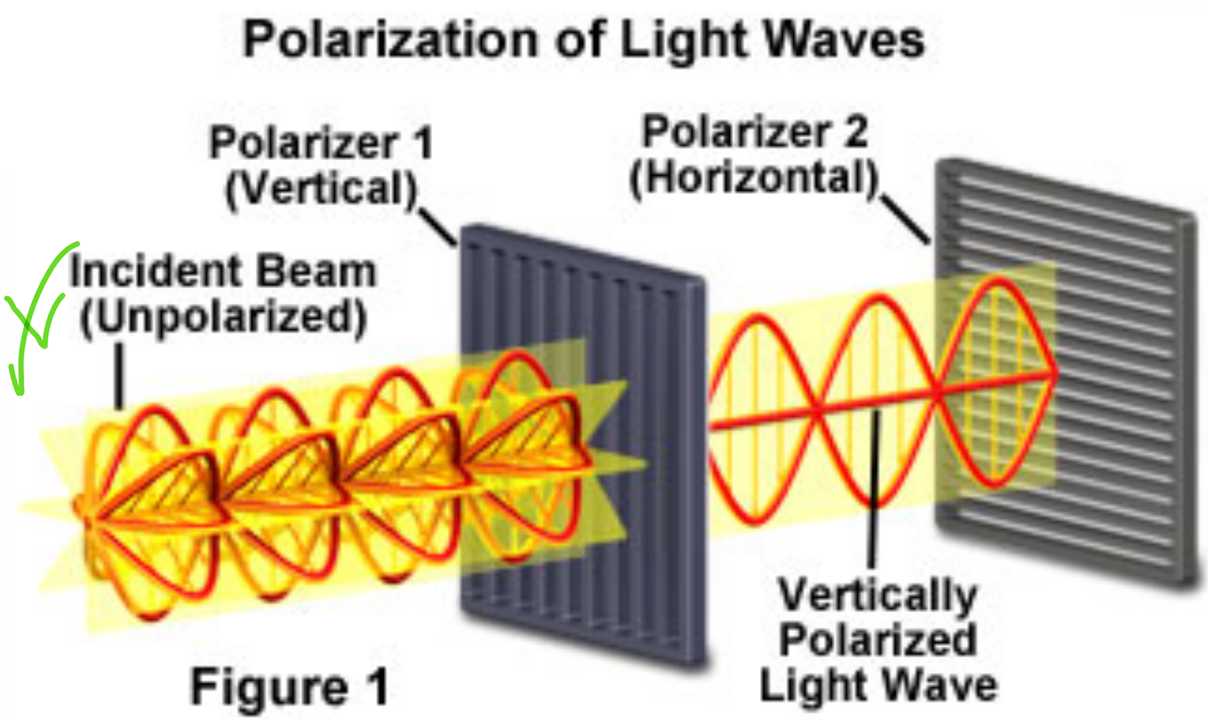
The electric field moves in one direction and the magnetic field in another ‘perpendicular to each other. So, we have one plane occupied by an electric field, another plane of the magnetic field perpendicular to it, and the direction of travel is perpendicular to both.

These electric and magnetic vibrations can occur in numerous planes.

A light wave that is vibrating in more than one plane is known as unpolarized light. The light emitted by the sun, by a lamp or a tube light are all unpolarized light sources.

Unpolarized and polarized light

In ordinary light the vibration of the electric field occurs in all directions in a plane Perpendicular to the direction of propagation.



If the electric field vectors are restricted to a single plane by filtration of the beam with specialized materials, then the light is referred to as **plane** or **linearly polarized** with respect to the direction of propagation, and all waves vibrating in a single plane are termed **plane parallel** or **plane-polarized.**

Methods Used in the Polarization of Light

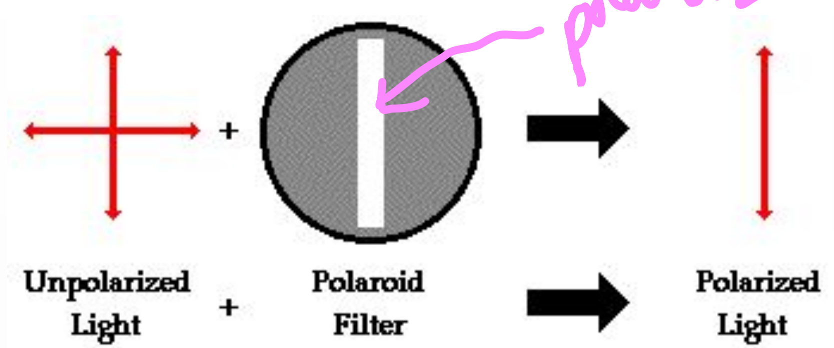
There are a few methods used in the polarization of light:

- Polarization by Transmission (polaroids)
- Polarization by Reflection
- Polarization by Scattering
- Polarization by Refraction

Polarization by Use of a Polaroid Filter

The most common method of polarization involves the use of a **Polaroid filter**. Polaroid filters are made of a special material that is capable of blocking one of the two planes of vibration of an electromagnetic wave. (Remember, the notion of two planes or directions of vibration is merely a simplification that helps us to visualize the wavelike nature of the electromagnetic wave.) In this sense, a Polaroid serves as a device that filters out one-half of the vibrations upon transmission of the light through the filter.

polarization axis



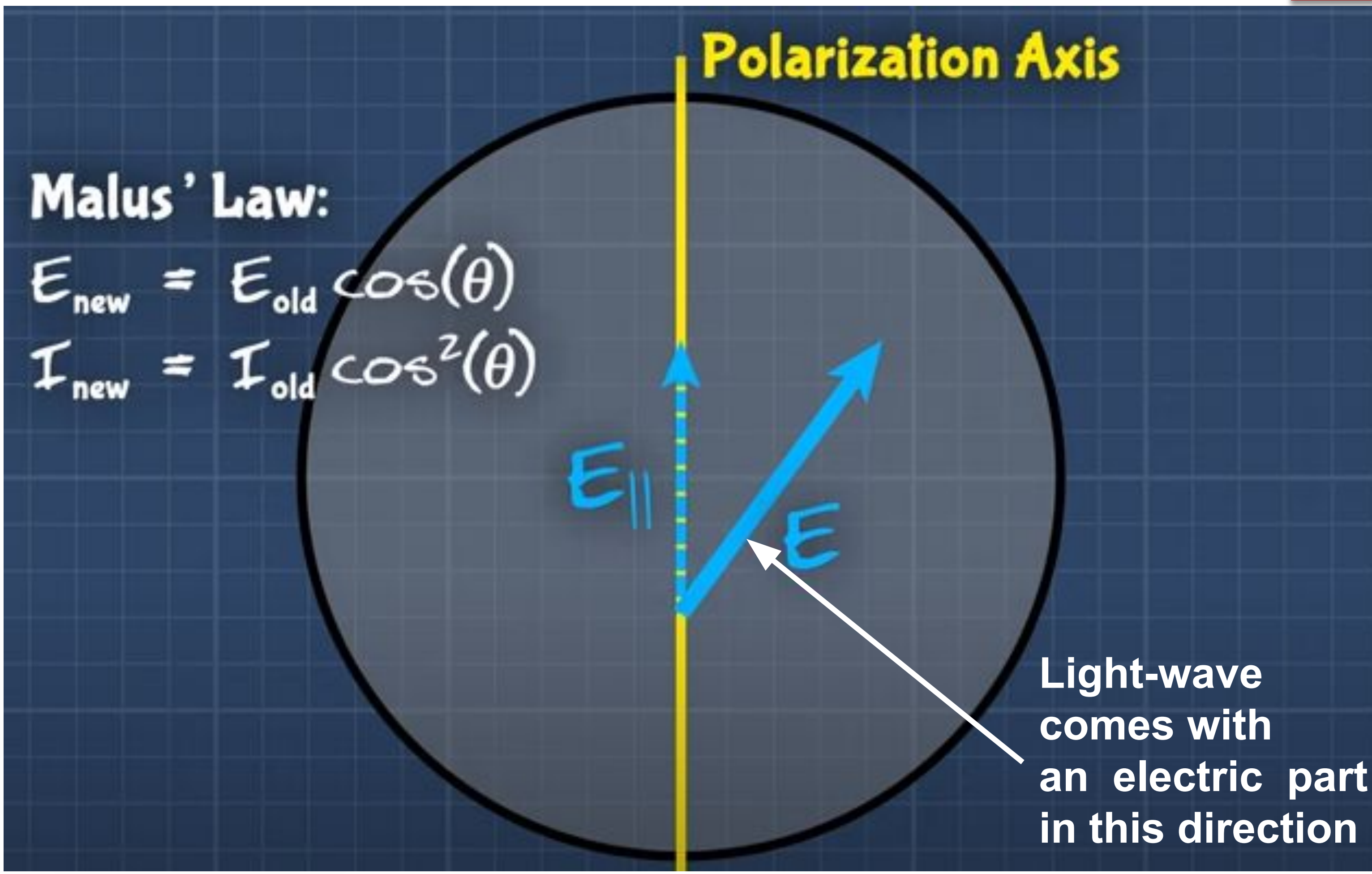
A Polaroid filter is able to polarize light because of the chemical composition of the filter material. The filter can be thought of as having long-chain molecules that are aligned within the filter in the same direction. During the fabrication of the filter, the long-chain molecules are stretched across the filter so that each molecule is (as much as possible) aligned in say the vertical direction. As unpolarized light strikes the filter, the portion of the waves vibrating in the vertical direction is absorbed by the filter. The general rule is that the electromagnetic vibrations that are in a direction parallel to the alignment of the molecules are absorbed.

The alignment of these molecules gives the filter a **polarization axis**. This polarization axis extends across the length of the filter and only allows vibrations of the electromagnetic wave that are parallel to the axis to pass through. Any vibrations that are perpendicular to the polarization axis are blocked by the filter. Thus, a Polaroid filter with its long-chain molecules aligned horizontally will have a polarization axis aligned vertically. Such a filter will block all horizontal vibrations and allow the vertical vibrations to be transmitted (see diagram above). On the other hand, a Polaroid filter with its long-chain molecules aligned vertically will have a polarization axis aligned horizontally; this filter will block all vertical vibrations and allow the horizontal vibrations to be transmitted.

Relationship Between Long-Chain Molecule Orientation and the Orientation of the Polarization Axis

When molecules in the filter are aligned vertically, the polarization axis is horizontal.

When molecules in the filter are aligned horizontally, the polarization axis is vertical.


Polarization Axis
Malus ' Law:
$E_{new} = E_{old} \cos(\theta)$
$I_{new} = I_{old} \cos^2(\theta)$
$E_{||}$
E
Light-wave
comes with
an electric part
in this direction

The following are the applications of polarization:

•Polaroid films are commonly used to reduce the glare of sunlight in bright settings. Since light reflected from water is partially polarized, these filters are used when photographing water bodies and when photographing objects underwater.

With filter

With filter

•Polaroid is used on spectacles to reduce glare. They are also used in night driving goggles as they can filter out the partially polarized light reflected off windshields.

•Three-dimensional movies are produced and shown with the help of polarization.

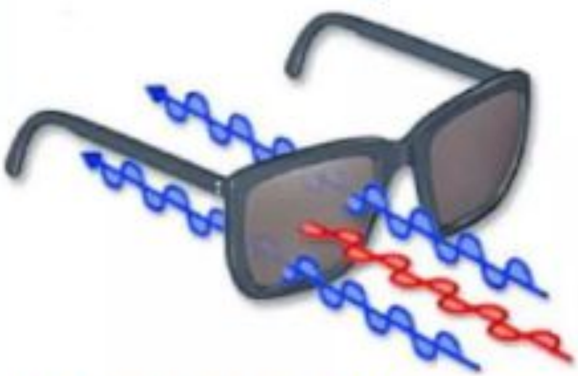
•Polarization is used for differentiating between transverse and longitudinal waves.

•Infrared spectroscopy uses polarization.

Action of Polaroid Sunglass

Light reflected from surfaces like a flat road or smooth water is generally horizontally polarized. This horizontally polarized light is blocked by the vertically oriented polarizers in the lenses.

Vertically Polarized Light from Objects

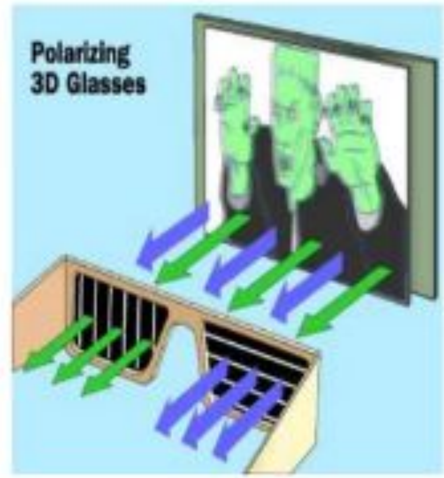


Unwanted glares are usually horizontally polarized light

https://www.youtube.com/watch?v=c9ew1J0PY-M

Polarization in 3D Movies

Polarization is often implemented in the production and viewing of 3D films. When watching a 3D movie, there are actually two images being projected onto the screen at once. The two images were filmed with two separate cameras from two slightly offset locations. These two images are projected onto the screen through Polaroid filters. The molecules of one of the filters is aligned vertically, the other is aligned horizontally. The audience is given 3D glasses that have one lens aligned horizontally and the other aligned vertically. Thus, one eye sees one image and the other eye sees the other image. The brain receives both signals and perceives depth on a flat screen.



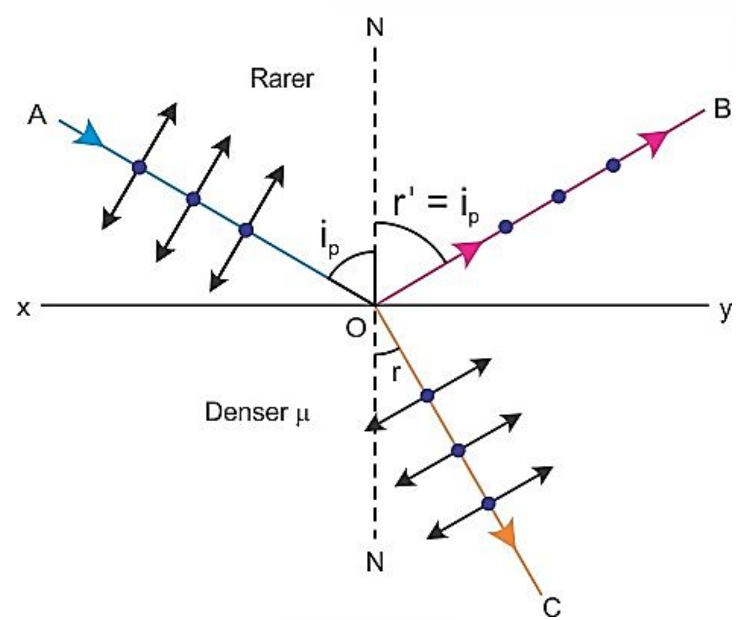
Polarization by reflection/ Brewster's law

- When light is reflected from a transparent medium (like glass) the reflected light is partially polarized in the horizontal direction.

- The degree of polarization depends on the angle of incidence.

- At a particular angle of incidence, when the reflected ray is perpendicular to the refracted ray then the reflected ray is plane polarized.

- The angle is called Brewster's angle or polarizing angle.



μ is the refractive index of the material
i is the polarization angle.

We can write from Snell's law,
$\mu = \sin i_p / \sin r$……………………(1)

The reflected light and refracted light should be perpendicular to each other.

Hence, we can write,
$i_p + r + 90 = 180$
$\Rightarrow r = 90 - i_p$

Hence, we can modify equation (1) such that,
$\mu = \sin i_p / \sin(90 - i_p)$
$\Rightarrow \mu = \sin i_p / \cos i_p$
$\Rightarrow \mu = \tan i_p$

This is the formula of Brewster's law.

Q1. The refractive index of water with respect to air is 4/3. Find the Brewster's angle for air-glass.

Q2. If a polarizing device exhibits a polarizing angle of 72°, what is the index of refraction of this medium?

Q3. Rays of light fall on a glass plate of refractive index $=\sqrt{3}$. If the angle between refracted ray and reflected ray is 90°, the angle of incidence?